1 **(a)** Calculates area of triangle: $\frac{1}{2} \times 12 \times 8 = 48$ **[M1]**

$4x = 48$

x = 12[$A1$]

(b) Uses Pythagoras to find slanted side: $\sqrt{6^2 + 8^2}$ **[M1]**

Slanted side = 10 **[A1]**

Perimeter = $12 + 10 + 10 = 32$ **[A1]**

[Total Marks: 5]

2 **(a)** $2(2x + 3) + 2(x - 1) = 34$ OR $2x + 3 + x - 1 = 17$ **[M1]**

$6x + 4 = 34$ OR $3x + 2 = 17$ **[M1]**

$x = 5$ **[A1]**

(b) Substitution of $x = 5$: $(2(5) + 3) \times (5 - 1)$ OR 13×4 **[M1]**

52 **[A1]**

[Total Marks: 5]

3 **(a)** $\frac{1}{2}(10 + 16) \times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle = $16 - 10 = 6$ **[M1]**

Slanted edge = $\sqrt{6^2 + 8^2} = 10$ **[M1]**

$8 + 10 + 16 + 10 = 44$ **[A1]**

[Total Marks: 5]

4 **(a)** Uses Area = $\frac{1}{2} \times$ base $\times$ height = $\frac{1}{2} \times 14 \times 9$ **[M1]**

63 **[A1]**

(b) Finds $AE = \sqrt{9^2 + 9^2} = \sqrt{162}$ or 12.7... **[M1]**

Finds $BE = \sqrt{5^2 + 9^2} = \sqrt{106}$ or 10.2... **[M1]**

Perimeter = $14 + \sqrt{162} + \sqrt{106}$ **[M1]**

37.0 (or 37.02...) **[A1]**

[Total Marks: 6]

5 **(a)** $\frac{1}{2} \times (15 + 24) \times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A = \sqrt{10^2 - 8^2} = 6$ **[M1]**

Horizontal distance for the right-angled triangle under $B = 24 - 15 - 6 = 3$ **[M1]**

$BC = \sqrt{8^2 + 3^2}$ $(= \sqrt{73} \approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Total Marks: 6]

6 **(a)** Area of rectangle $= 4(2x + 3)$ OR $8x + 12$ **[M1]**

Area of triangle $= \frac{1}{2} \times 6 \times (3x - 1)$ OR $9x - 3$ **[M1]**

$8x + 12 = 9x - 3$ **[M1]**

$x = 15$ **[A1]**

(b) Length $= 2(15) + 3 = 33$ **[M1]**

Perimeter $= 2(4 + 33) = 74$ **[A1]**

[Total Marks: 6]

7 **(a)** **(i)** Sets up equation for area: $\frac{1}{2}((x + 4) + (3x - 2))(x - 1) = 20$ **[M1]**

Simplifies expression to $\frac{1}{2}(4x + 2)(x - 1) = 20$ or $(2x + 1)(x - 1) = 20$ **[M1]**

Expands to $2x^2 - 2x + x - 1 = 20$ leading to $2x^2 - x - 21 = 0$ with no errors seen **[A1]**

(ii) Factorises $(2x - 7)(x + 3) = 0$ or uses quadratic formula to find $x = 3.5$ **[M1]**

Calculates straight sides: $AD = 7.5$, $BC = 8.5$, $AB = 2.5$ **[M1]**

Sets up Pythagoras for the slanted side CD: $\sqrt{(8.5 - 7.5)^2 + 2.5^2}$ **[M1]**

Calculates $CD = \sqrt{1^2 + 2.5^2} = \sqrt{7.25}$ (or $2.69\ldots$) **[A1]**

Perimeter $= 7.5 + 8.5 + 2.5 + 2.69\cdots = 21.2$ (or $21.19\ldots$) **[A1]**

[Total Marks: 8]

8 **(a)** $\frac{1}{2}(x + x + 4)(x - 1) = 40$ **[M1]**

$\frac{1}{2}(2x + 4)(x - 1) = 40$ OR $(x + 2)(x - 1) = 40$ **[M1]**

$x^2 + 2x - x - 2 = 40 \implies x^2 + x - 42 = 0$ **[A1]**

(b) $(x + 7)(x - 6) = 0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top = 6, Bottom = 10, Height = 5 **[M1]**

Horizontal difference = 4, so slanted side $= \sqrt{5^2 + 4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter $= 5 + 6 + 10 + \sqrt{41}$ **[M1]**

27.4 (or $27.40\ldots$) **[A1]**

[Total Marks: 9]

9 **(a)** Perimeter of rectangle $= 6y$ OR Perimeter of square $= 60 - 6y$ **[M1]**

$\frac{60-6y}{4}$ OR $15 - 1.5y$ **[A1]**

(b) $(15 - 1.5y)^2 + y(2y) = 113$ **[M1]**

$225 - 45y + 2.25y^2 + 2y^2 = 113$ **[M1]**

$4.25y^2 - 45y + 112 = 0$ leading to $17y^2 - 180y + 448 = 0$ with no errors seen **[A1]**

(c) $\frac{-(-180) \pm \sqrt{(-180)^2 - 4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180 \pm \sqrt{1936}}{34}$ OR $\frac{180 \pm 44}{34}$ **[M1]**

$y = 4$ and $y = 6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

(d) Identifies $y = 4$ as the valid solution (since $15 - 1.5(6.59) < 6.59$) leading to side length $= 9$ **[M1]**

81 **[A1]**

[Total Marks: 10]